Change of Coordinates, $\vec{\S}4.6$

Recall from last class:

- If V is spanned by a finite set, then V is called **finite dimensional,** and the **dimension** of V, written $\dim(V)$, is the number of vectors in a basis of V. Define $\dim(\{0_V\}) = 0$. If V is not spanned by any finite set, then V is called **infinite dimensional**.
- **Theorem:** Let H be a subspace of a finite dimensional vector space V. Any linearly independent set in H can be expanded to a basis for H, and H is finite dimensional and $\dim(H) \leq \dim(V)$.
- **Basis Theorem:** Let V be an n-dimensional vector space with $n \geq 1$. Any linearly independent set of exactly n vectors in V is a basis for V. Any set of exactly n vectors that spans V is a basis for V.
- Let A be an $m \times n$ matrix with rows $a_1, \ldots, a_m$. Define the **row space** of A by
$$\text{Row}(A) = \text{Span}\{a_1, \ldots, a_m\}.$$
- **Theorem:** If A is row equivalent to B, then $\text{Row}(A) = \text{Row}(B)$. Furthermore, if B is in reduced row echelon form, the non-zero rows of B form a basis for $\text{Row}(A)$ and $\text{Row}(B)$.
- The **rank** of the $m \times n$ matrix A is the dimension of $\text{Col}(A)$. We write $\text{rank}(A) = \dim(\text{Col}(A))$. The **nullity** of A is the dimension of $N(A)$.
- **Rank-Nullity Theorem:** Let A be an $m \times n$ matrix.
  1. $$\dim \text{Row}(A) = \dim \text{Col}(A)$$
  2. $$\text{rank(A)} + \dim N(A) = n$$
- **Invertible Matrix Theorem Continued** The following are equivalent:
  - m. The columns of A form a basis of $\mathbb{R}^n$.
  - n. $\text{Col}(A) = \mathbb{R}^n$.
  - o. $\dim \text{Col}(A) = n$.
  - p. $\text{rank}(A) = n$.
  - q. $N(A) = \{0\}$.
  - r. $\dim N(A) = 0$.
- **Theorem:** Let $T : V \to W$ be a linear transformation between vector spaces. If V is finite dimensional, then
$$\dim \text{im}(T) + \dim \ker(T) = \dim(V).$$

Rank-Nullity Theorem: Let A be an $m \times n$ matrix.

a.
$$\dim \text{Row}(A) = \dim \text{Col}(A)$$

b.
$$\text{rank(A)} + \dim N(A) = n$$

1.

(a) Let A be a 7 by 5 matrix. What is the largest possible rank?

(b) Let A be a 3 by 7 matrix. What is the smallest possible nullity?

(c) Let A be a 5 by 6 matrix. Suppose the solutions to $Ax = 0$ are all multiple of one non-zero vector. Will the matrix equation $Ax = b$ always be consistent?

Invertible Matrix Theorem Continued The following are equivalent to A being invertible:

m. The columns of A form a basis of $\mathbb{R}^n$.

n. $\text{Col}(A) = \mathbb{R}^n$.

o. $\dim \text{Col}(A) = n$.

p. $\text{rank}(A) = n$.

q. $N(A) = \{0\}$.

r. $\dim N(A) = 0$.

Theorem: Let $T : V \to W$ be a linear transformation between vector spaces. If V is finite dimensional, then
$$\dim \text{im}(T) + \dim \ker(T) = \dim(V).$$

2. Let $\mathcal{B} = \{b_1, b_2\}$ and $\mathcal{C} = \{c_1, c_2\}$ be bases for a vector space V where $b_1 = c_1 + c_2$ and $b_2 = -c_1 + c_2$. How can we change from the basis $\mathcal{B}$ to $\mathcal{C}$?

Theorem: Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ and $\mathcal{C} = \{c_1, \ldots, c_n\}$ be bases of a vector space V. Then there exists a unique $n \times n$ matrix P such that $[x]_{\mathcal{C}} = P[x]_{\mathcal{B}}$.

3. Prove the theorem.

Note: We call P the **change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$**. In general

$$P = [[b_1]_{\mathcal{C}} \ \dots \ [b_n]_{\mathcal{C}}].$$

Also P^{-1} is the change of coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

4. Let $b_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $b_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $c_1 = \begin{bmatrix} -7 \\ 9 \end{bmatrix}$, and $c_2 = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$. Find the change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$.

5. Let $\mathcal{B} = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$ and $\mathcal{E} = \{1, t, t^2\}$ be bases for $\mathbb{P}_2$. Find the change of coordinates matrix from $\mathcal{B}$ to $\mathcal{E}$.

6. Find $[-1 + 2t]_{\mathcal{B}}$.

Let V be a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and W be a vector space with basis $\mathcal{C} = \{c_1, \dots, c_m\}$. Given a linear transformation $T : V \to W$, we naturally seek an associated matrix to represent T using the bases; namely we want an $m \times n$ matrix M such that $[T(x)]_{\mathcal{C}} = M[x]_{\mathcal{B}}$ for all $x \in V$. We call M the **matrix of T relative to bases $\mathcal{B}$ and $\mathcal{C}$**. In general $M = [[T(b_1)]_{\mathcal{C}} \ \dots \ [T(b_n)]_{\mathcal{C}}]$.

7. Let $\mathcal{B} = \{b_1, b_2, b_3\}$ and $\mathcal{D} = \{d_1, d_2\}$ be bases for V and W respectively. Let $T : V \to W$ be the linear transformation such that $T(b_1) = 3d_1 - 5d_2$, $T(b_2) = -d_1 + 6d_2$, and $T(b_3) = 4d_2$. Find the matrix of T relative to $\mathcal{B}$ and $\mathcal{D}$.

Change of Coordinates – Answers / Solutions

1. Since we are given b_1 as a linear combination of c_1 and c_2, we can give the coordinates of b_1 in terms of basis $\mathcal{C}$; i.e.

$$[b_1]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Similarly, we get

$$[b_2]_{\mathcal{C}} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

. Let $x \in V$, then $x = c_1 b_1 + c_2 b_2$ for some scalars c_1 and c_2. Thus

$$[x]_{\mathcal{C}} = [c_1 b_1 + c_2 b_2]_{\mathcal{C}}.$$

Since the coordinate mapping is linear, we get

$$[x]_{\mathcal{C}} = c_1[b_1]_{\mathcal{C}} + c_2[b_2]_{\mathcal{C}} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} [x]_{\mathcal{B}}.$$

Thus we can change form basis $\mathcal{B}$ to $\mathcal{C}$ by multiplying by the matrix $P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

2. For the proof, we carry out the strategy in problem 1 more generally.

Proof. Let $x \in V$, then there exists $d_1, \dots, d_n \in \mathbb{R}$ such that

$$[x]_{\mathcal{B}} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix}.$$

Then using the linearity of the coordinate mapping and the definition of matrix multiplication, we get

$$\begin{aligned} [x]_{\mathcal{C}} &= [d_1 b_1 + \cdots + d_n b_n]_{\mathcal{C}} \\ &= d_1[b_1]_{\mathcal{C}} + \cdots + d_n[b_n]_{\mathcal{C}} \\ &= [[b_1]_{\mathcal{C}} \ \dots \ [b_n]_{\mathcal{C}}] \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} \\ &= P[x]_{\mathcal{B}}. \end{aligned}$$

Thus the matrix P is the change of coordinates matrix from basis $\mathcal{B}$ to basis $\mathcal{C}$. $\square$

3. Recall the matrix P that changes coordinates from basis $\mathcal{B}$ to basis $\mathcal{C}$ is given by

$$P = [[b_1]_{\mathcal{C}}; [b_2]_{\mathcal{C}}].$$

Thus we want to find scalars a_1, a_2 such that $b_1 = a_1c_1 + a_2c_2$. Thus we want to solve the matrix equation $[c_1\ c_2]\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = b_1$. We could row reduce the augmented matrix $[c_1\ c_2|b_1]$ or we could use $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = [c_1\ c_2]^{-1}b_1$. Using the inverse matrix, we get

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -7 & -5 \\ 9 & 7 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \frac{1}{-4}\begin{bmatrix} 7 & 5 \\ -9 & -7 \end{bmatrix}\begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Thus $[b_1]_{\mathcal{C}} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

Similarly, we have $b_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} -7 & -5 \\ 9 & 7 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$. Again solving the matrix equation, we get $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \frac{-3}{2} \\ \frac{5}{2} \end{bmatrix}$. Thus $[b_2]_{\mathcal{C}} = \begin{bmatrix} \frac{-3}{2} \\ \frac{5}{2} \end{bmatrix}$.

Thus

$$P = \begin{bmatrix} 2 & -\frac{3}{2} \\ -3 & \frac{5}{2} \end{bmatrix}.$$

Equivalently we could use that P is equal to first changing coordinates from $\mathcal{B}$ to the standard coordinates (given by $P_{\mathcal{B}}$) and then changing from the standard coordinates to $\mathcal{C}$ (given by $P_{\mathcal{C}}^{-1}$). Thus

$$P = P_{\mathcal{C}}^{-1}P_{\mathcal{B}} = \begin{bmatrix} -7 & -5 \\ 9 & 7 \end{bmatrix}^{-1}\begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{3}{2} \\ -3 & \frac{5}{2} \end{bmatrix}.$$

4. Let P denote the change of coordinates matrix from $\mathcal{B}$ to the standard coordinates, then

$$P = \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix}.$$

5. Recall

$$[x]_{\mathcal{E}} = P[x]_{\mathcal{B}}.$$

We want to solve the matrix equation $P[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$. Solving the augmented matrix we get

$$\left(\begin{array}{ccc|c} 1 & 3 & 0 & -1 \\ -2 & -5 & 2 & 2 \\ 1 & 4 & 3 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right).$$

Thus $[-1+2t]_{\mathcal{B}} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$, and hence

$$x = 5(1 - 2t + t^2) - 2(3 - 5t + 4t^2) + 1 \cdot (2t + 3t^2).$$

6. Recall the matrix M of T relative to bases $\mathcal{B}$ and $\mathcal{D}$ is given by $M = [[T(b_1)]_{\mathcal{D}}\ [T(b_2)]_{\mathcal{D}}\ [T(b_3)]_{\mathcal{D}}]$. Since $T(b_1) = 3d_1 - 5d_2$, we get $[T(b_1)]_{\mathcal{D}} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$. Since $T(b_2) = -d_1 + 6d_2$, we get $[T(b_2)]_{\mathcal{D}} = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$. Since $T(b_3) = 4d_2$, we get $[T(b_3)]_{\mathcal{D}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Thus the matrix of T relative to bases $\mathcal{B}$ and $\mathcal{D}$ is given by $\begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix}$.